

SPEEDING THROUGH THE FRAT HOUSE: A QUALITATIVE EXPLORATION OF NONMEDICAL ADHD STIMULANT USE IN FRATERNITIES

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ABSTRACT

Qualitative methods were used to investigate the use of nonmedical Attention Deficit Hyperactivity Disorder (ADHD) stimulants by fraternity members. The primary goal of the study was to determine students' levels of understanding and motivations for use of these Schedule II controlled substances. Seventy-nine in-depth interviews were conducted. Key findings highlighted how: a) easy it was for subjects to obtain stimulants; b) little health information nonmedical users have about stimulants; c) academic stress created a fertile context for stimulant use; and d) a small number of prescribed users supply the vast majority of nonmedical users with their stimulants. Most nonmedical users claimed to primarily use ADHD stimulants in periods of high academic stress and believed that they not only reduced fatigue, but also increased reading comprehension, interest, cognition, and memory. These qualitative data have supplied a rich and complex understanding behind nonmedical ADHD stimulant use among fraternity members.

INTRODUCTION

In recent years, researchers have begun to identify the number of college students obtaining and using ADHD stimulants on American college campuses (McCabe,

Teter, & Boyd, 2004, 2006; Teter, McCabe, LaGrange, Cranford, & Boyd, 2006; White, Becker-Blease, & Grace-Bishop, 2006). They have discovered while Ritalin (Methylphenidate), and Dexedrine (dextroamphetamine) are occasionally used, Adderall (mixed salts amphetamine) is the overwhelming drug of choice. Because of the potential for abuse and psychological and physical dependency, the U.S. Drug Enforcement Administration (DEA) has classified these stimulants as Schedule II substances¹ (Woodworth, 2000).

Despite this classification, several studies have documented the nonmedical use of ADHD stimulants on college campuses. Precise estimates of the prevalence of nonmedical use on college campuses, however, have tended to vary. McCabe, Knight, Teter, and Wechsler (2005) in a multi-site study surveyed 10,904 students at 119 nationally representative 4-year colleges in the United States. They found 6.9% of the students surveyed had used a prescription stimulant in the last year, and reported nonmedical use was highest among

1. students from the Northeastern region of the United States; and
2. students from colleges with more competitive admission standards.

Other investigations have yielded values higher than McCabe et al. (2005). For instance, Hall, Irwin, Bowman, Frankenberger, and Jewett (2005) reported 17% of males and 11% of females surveyed used stimulant medication without a prescription during their lifetime. Babcock and Byrne (2000) found that 16% of their sample used ADHD stimulants recreationally during their lifetime. DeSantis, Webb, and Noar (2008) found 34% of 1,811 students surveyed at a large public institution in the American Southeast claimed to have ever used ADHD medication nonmedically (lifetime use). Finally, examining use of amphetamines in a convenience sample of students, Low and Gendaszek (2002) found a prevalence estimate of 35.5% for past year use.

Studies have also found college students use ADHD medications without prescriptions for both academic and recreational purposes. For example, Teter et al. (2006) found participants ($N = 382$) primary motives for use were to:

1. “help with concentration” (65.2%);
2. “help study” (59.8%); and
3. “increase alertness” (47.5%).

¹According to The Federal Drug Administration’s (FDA) medication guides for commonly prescribed ADHD stimulants (Adderall and Ritalin), common side effects of the drugs include headache, stomach upset, nausea, loss of appetite, trouble sleeping, weight loss, dry mouth, nervousness, mood swings, dizziness, fast heart beat, and heart palpitations (Medication Guide Adderall XR, 2007; Medication Guide Ritalin, n.d.). These guides also highlight that these medications may pose a risk for individuals with preexisting conditions such as heart problems, heart defects, high blood pressure, liver or kidney problems, thyroid problems, and a history of seizures.

In this study 31% of nonmedical users also reported using stimulants to get high. In another study of undergraduates, Teter, McCabe, Cranford, Boyd, and Guthrie (2005) found nonmedical users ($N = 689$) reported using prescription stimulants to:

1. help concentration (58%);
2. increase alertness (43%); and
3. obtain a high (43%) as their primary motivations for use.

While these data give us some understanding of the prevalence of nonmedical stimulant use and provide some basic answers as to why fraternity members use these substances, they stop short of supplying a thick description and complex understanding of the behaviors, conceptions, and choices of users that can only be mined through qualitative methods. Fraternity members are a particularly important group to study as several studies have found higher levels of stimulant use among this population (DeSantis et al., 2008; McCabe et al., 2005, 2006; Shillington, Reed, Lange, Clapp, & Henry, 2006). Consequently, the purpose of this research is to investigate how and why fraternity members, at an American public university, use nonmedical prescription stimulants through the use of both quantitative survey data and in-depth qualitative interviews. Specifically, we examine:

1. factors that led to first use of prescription ADHD medications;
2. motives for continued use of these medications; and
3. where and how students access these medications.

METHODS

This study utilized in-depth qualitative interviews. The university's Institutional Review Board Human Subjects Committee approved all methods for the current study.

Qualitative Interviews

Seventy-nine full-time male fraternity undergraduates at a large, public southeastern research university in the United States were interviewed using hand-held audio tape recorders. The first 63 of these interviews were conducted during Spring and Summer of 2006 as part of a broader study (DeSantis et al., 2008). An additional 16 interviews were conducted during the Summer of 2008 in order to provide additional data for this in-depth study examining only fraternity members. Each interview lasted an average of 30 minutes.

Subjects were selected by convenience. The researchers, however, made an effort to interview fraternity members representing as many different organizations as possible and to avoid over-reliance on any single organization. Because of IRB restrictions, the names and specific organizations of our subjects could not be recorded or cataloged.

Along with the primary author of this study, two paid undergraduate students also conducted interviews. The undergraduate interviewers had all completed an upper-division research methods class and were certified by the university's Institutional Review Board. Additionally, the interviewers were trained and supervised by the researchers in interviewing procedures, ethical guidelines, and transcription protocol. Each undergraduate interviewer was given a detailed interviewing script to follow and obtained written consent before each interview. Also, audio interviews were transcribed periodically throughout the study and evaluated by the study authors in order to maintain consistency among the interviewers.

During the transcription stage of this project, all names were changed and replaced with arbitrary pseudonyms to protect the anonymity of the subjects. Furthermore, any identifying markers and/or references to people, organizations, or events that could jeopardize subject anonymity were either changed or deleted from the transcription record.

The interviewing script focused on five major areas of nonmedical stimulant use:

1. first-time use;
2. motivations for use;
3. perceived advantages;
4. perceived side-effects; and
5. availability and procurement.

The foundation for the interviewing guide came from previous quantitative research examining nonmedical use of ADHD stimulants (McCabe et al., 2004, 2005; Teter et al., 2005). Questions were also crafted in order to explore areas that had previously been unexplored, such as the context of stimulant use and beliefs about use. In order to protect the anonymity of the subjects, all names were changed and replaced with arbitrary pseudonyms during transcription. All identifying markers and/or references to organizations, events, and people were also either changed or deleted from the transcriptions.

The five major areas of the interviewing script (listed above) served as an initial organizing framework for the findings. Two independent coders conducted several close readings of the focus group transcripts in order to identify themes. The coders then met to discuss both general themes and more specific categories. The coders met to establish consensus and to clarify themes.

RESULTS

First-Time Use

Before conducting our interviews, DeSantis, Noar, and Webb's (2009) survey research of $N = 333$ fraternity members supplied us with statistical data of how prevalent illegal stimulant use was on this campus. They found that at least 65% of the nonmedical users in the study first used prescription stimulants in

college. This statistic only reveals a small part of the picture about first-time use for fraternity members. The qualitative interviews provided an opportunity to explore this experience in more depth.

The interviews revealed casual discussions of nonmedical stimulant use occur frequently among fraternity members. ADHD stimulants were a salient part of fraternity life and the larger university culture. Thus, first-time users had no need to seek out information about the drug.

John remembers after arriving on campus, "Everybody was always talking about it, in high school too. But it was just really common." Brian laughed about the prevalence of stimulants: Ha, it was crazy. I'm from a small town, so I was not really sure a lot about it. It was big here. Especially, during finals, as if it was just a Coke, no big deal.

While discussions about nonmedical prescription stimulant use were common on campus, these discussions became even more prevalent inside fraternities. "So I moved into the house," Brian, recalls, "and it seemed like everyone was on it. Sometime, everyone was on it [laughter]. It was something that no one like hid at all." Jason speculated, "Since everyone lives together, it is just a lot more popular with fraternities . . . it is always around, in my house at least. No real big deal." Robert concurs: "I think fraternity use is much higher . . . I mean, you're all brothers and don't have to worry about anybody. So yes, it's just like, 'matter of fact.'"

Most students also agreed first-time use was engendered by periods of high academic stress and anxiety. "The first time I took it," remembers Adam, "was a really busy week. I didn't think that I could get through it. So my roommate had some and we did it. He didn't even need it [laughter]. It worked. I did good." For Kevin, "It was crazy. I had like three things due and I was just spent. So I decided to do it just to help get through. It was either that or crash and burn."

The pressure-filled nature of finals created the ideal situation for nonmedical stimulant use for over two-thirds of interviewees. "It's always finals I think," remarked Kyle. "That was mine. You just have to keep running like a marathon." For Ben, it was "finals" and "The fact that the whole house was doing it. It was like everyone just did it to get through. I mean, the week just really sucks."

The added stress taking multiple final exams on the same day also leads many to use Adderall. "It's the days when you have two or three tests that are the worst," explained Eric. "There is not enough time to do everything they make you. To be honest, sometimes there is no other way to do it all." For Jason, it is the only time he uses nonmedical stimulants: "I don't need it 99% of the time. I, for the most part, you know, can handle school."

Finally, the only knowledge these students had about ADHD medications was the praise they heard from other students about the drugs' effectiveness. John heard it was a "miracle!" and a "study drug" from his fraternity brothers. Chris heard that it was "the best thing for studying" and that "you will do better in school." For Martin and Ben, it was the repeated testimony from older brothers

who “never felt bad,” “made it through finals,” and “got better grades” because of it. None of the 79 subjects that we interviewed, however, sought out information from health professionals, medical or pharmaceutical reference guides, or even Internet sites before taking their first dose.

Motivations for Taking Nonmedical ADHD Medications

Academic Motives

Unlike most other substances, these fraternity members did not primarily take prescription stimulants for recreation. Alcohol, marijuana, prescription barbiturates, and cocaine were used almost exclusively to “get high” and to “have fun,” but ADHD medications were primarily used for “getting good grades.” DeSantis et al. (2009) found that 74% of nonmedical users reported using to study longer. “You know how tiring finals are?” remarked Anthony. “You are just running on fumes. It’s either that [taking stimulants], or you sleep and you can’t do that.” Procrastination, not the demands of finals, leads Jeremy to use: “It’s my own fault most of the time. I know I should start earlier, but I don’t so I end up cramming all the time. So I have to stay up.”

The students’ use of amphetamines to stay awake was not entirely surprising. Fighting fatigue and increasing energy are the most obvious and well-known effects of stimulants. As Justin, an Alpha brother, told us, “That’s why they call it speed.” Interestingly, many nonmedical users touted ADHD medications for their ability to improve concentration and focus. Fifty-nine percent of non-medical users reported taking prescription stimulants “to help concentrate” on schoolwork. Below the surface of the statistical data, the students’ conception of “concentrating” has multiple and layered meanings. Specifically, these students claim that they can a) study longer, b) stay focused without distraction, and c) be more productive.

For many students, the most important part of concentrating was the ability to study for longer periods of time. “Sometimes when I cram, explains Kyle, I need to study for a long period of time and it really helps with that.” Similarly, Thomas believes that unlike his usual pattern of studying for an hour before needing a break, he “. . . can sit for like 4 hours and just push through the stuff.” For Jason, “It’s just the easiest way to cram. I have trouble sitting still, and it [Adderall] just keeps me focused when I need it.”

Other students conceptualized the ability to focus without distraction as “concentrating.” For example, Mark says that it “blocks out the outside noise” that he “personally pick[s] up on.” Pete claimed stimulants help “block everybody out around” him. “I get really into the details . . . and I can concentrate like a laser on my organic chemistry, or whatever.” Many participants commonly defined this cognitive effect as “being in the zone.”

Most of the participants also claimed stimulants increased productivity. "Normally, it takes me a while to do things," explains Paul. "But when I take Adderall, I am really productive. I can get so much done, like a week's worth of stuff in like a day." Cameron explained, he is also "so much more productive. I mean I'm generally productive. It's just like a different level on Adderall."

Students also reported perceptions of increased intelligence and heightened cognitive aptitude as a motivation. As Mitch succinctly remarked, "the stuff is like an academic anabolic steroid." This "steroid effect" can best be understood as the enhanced ability to memorize and grasp ideas. For example, DeSantis et al. (2009) found that 30% of participants equated "being smarter" to the heightened facility to memorize and retain information. "I'm telling you, really, I can remember more. It just stays in and it's easier," confessed Kyle. Travis similarly believes when he takes Adderall, his "mind soaks up the material better. . . . I am reading the same thing, but it gets soaked in on one time so I got it a lot better; Does that make sense? It's easier to memorize anything."

Others, like Justin, an economics major, claim stimulants dramatically increase the ability to "grasp ideas" that would "normally be too hard to get." He continues: "Some of the stuff we have to know is really hard for me to get. I am not good at math. So it will be during finals and I'll take 20 milligrams and somebody is explaining it to me, and I get it. It always happens." Similarly, Adam stated: "Instead of just memorizing financial [accounting], something like that, it makes sense to me. I understand why I'm doing it. I mean, not all the time, but yeah, I think I am a little smarter on it."

Finally, DeSantis et al. (2009) found that 9% of users reported taking stimulants to make academic work more interesting. "I can get into school work more," reported Robert. "I have to be honest; it is one of the reasons that I really love this stuff [Adderall]. No matter what it is, it is just more interesting. Like biology, last semester, I remember being really into it." Sean claims everything is more interesting: "Really, TV, ESPN, cleaning your room [laughing], reading. That's the great thing about it." One might assume nonmedical users with academic motivations are better students, but this is not the case. Data indicate that users are more likely to have *lower* GPAs when compared to non-users (DeSantis et al., 2009).

In summary, our data revealed the subtle motives defining when and why these students take stimulants. For example, our subjects primarily take stimulants during finals week or periods of high academic stress. These same students take stimulants to:

1. stay awake;
2. concentrate (e.g., focus longer, avoid distraction, be more productive);
3. increase cognitive aptitude (e.g., aid memorization, comprehension); and
4. make material more engaging. But academic improvement was not the only motive for nonmedical stimulant use.

In the following section, we investigate the non-academic motives for taking ADHD medications.

Non-Academic Motives

The data demonstrate students primarily use stimulants to enhance academic performance. Some Greek students, however, turn to stimulants for recreation. DeSantis et al. (2009) found that, of the students who reported nonmedical use, 17% also reported taking them “to stay awake and have fun” and 3% claimed using for “the high.” Beyond these numbers, interviewees detailed a number of ways stimulants are used to promote “fun.”

Given the highly social nature of Greek life, it is not surprising the primary social advantage for stimulant use is the additional energy it supplies to users. “Partying is draining,” remarked Jake. “If you have a big night and you are tired, you just take Adderall and you are set to go.” Kevin claims, Saturdays during football season are “prime time” for use. “It is a long day usually. We start early and if it is a day game, and you have to keep partying after, you are dead. You can’t do it without help [laughter]. Long days and nights was also Justin’s motivation: “If I start drinking early, I am dead by eight. . . . Ten milligrams and you can have a great night and be ready to go until late.”

Along with staying awake, participants also claimed stimulants make them more social and talkative. “Everybody” tells Scott, for example, he is “more outgoing and funnier” on Adderall. “They all say that I am a lot more extroverted and funny, really.” For Ryan, Adderall makes him feel more social: “I just feel like I am ‘on’ and more relaxed. Hanging out and talking is easy, to girls. I probably am talking too much or like an idiot. But I feel good.”

Other students sought the interactive effects of combining alcohol and stimulants. Interestingly, our interviewees could not agree on what specifically happens when you combine alcohol and stimulants. Zach claimed “taking one before you go out drinking gets you drunker, quicker.” Mark also asserted, “It is great for partying. You get fucked up faster and you can keep going without passing out.” For others, however, Adderall was used to decrease the effects of alcohol. “If you have a long night, it seems to keep you under control more,” explained Chris. “You don’t seem to get fucked up so fast, or sloppy . . . which is good sometimes.”

A small number of subjects, however, preferred the unaltered effects of stimulants. Three percent of users admitted to taking stimulants because they are a cheap, easily accessible high. “It’s kinda like coke,” explained David. “You just crush it up and snort it.” Supplying more detail, Eric tells how “Adderall is time released for like 12 hours or something. So when you smash it up with a spoon, you get it all at once so it is a stronger rush.” “The best thing” about snorting ADHD stimulants, concludes Matthew, “is that it is cheap coke. You know you get that same high but you can get it for like 5 bucks.”

Many of the subjects in our study found utility in Adderall's ability to fight fatigue (especially during extended periods of partying), increase sociability, heighten the effects of alcohol, decrease the depressant qualities of alcohol, and simulate cocaine. The obvious question left unanswered is how so many students obtain such a popular, versatile, and highly controlled medication.

Obtaining Stimulants

The ADHD stimulants investigated in this study are classified by the DEA as Schedule II drugs because of their high potential for nonmedical use and risk of severe psychological and/or physical dependence (USDOJ DEA). Consequently, Adderall, Ritalin, Dexedrine, and the like are legally available only through prescription, with a limit of 30 days' worth of doses, and no refills. Additionally Schedule II drugs are subject to production quotas set by the DEA.

For most of the students in our study the DEA's Schedule II classification means little. "Really, the stuff is everywhere," remarked Tony. "It's no big deal to find it at all." Other students agreed. DeSantis et al. (2009) reported that, when asked how difficult it is to obtain the stimulants, students reported: "very easy" (47%), "somewhat easy" (43%), and "somewhat difficult" (9%). Only one of the subjects thought obtaining stimulants was "very difficult." Since 90% of the participants in DeSantis et al. (2009) study thought obtaining stimulants was either "very" or "somewhat easy," we asked our subjects to expand on where and how they get their drugs.

The students in their study did not get their stimulants from the typical (or stereotypical) "drug dealer." Instead, the vast majority of users claimed to get their substances from "friends" (DeSantis et al., 2009). Only 9% claimed they procure their stimulants from "strangers." Only 4% in the DeSantis et al. (2008) campus population study claimed to have a prescription from a doctor to treat their diagnosed ADHD. Apparently, this 4% is a primary distribution source to the Greek population.

For fraternity members, the 4% of students with prescriptions are perceived to be "everywhere." "In our house, there is always a few brothers with it," explained Eric. "So we really turn to them, especially during finals. No problem. They are really good about hooking us up first. It's easy. Except during finals when we all want it [laughter]."

Three aspects of these student's personal relationships facilitates the ease of procuring stimulants. First, many brothers share living space with each other, often with little privacy. "In our house, everybody is in everybody's business. There is just too many of us crammed in to have secrets." Greg explained in such a context, "Everybody knows who has the scripts. We are all brothers."

Second, because these students are "brothers," with all the familial connotations attached to that bond, there is little fear asking for, or doling out, prescription medications could have any negative consequences. "It's real safe for me,"

explained Travis. "I get mine from my brothers so I never really worry about it." Scott agreed: "I know they are good. I mean that's kinda the thing. They come from a [pharmaceutical prescription] bottle from someone you know. It's not like you get it from pushers standing on a corner, or anything like that."

Finally, given the highly social nature of Greek organizations, fraternity men have a large social network of friends. For example, the 42 fraternities (with an average of approximately 70 members) and sororities (with an average of approximately 130 members) on campus may sponsor two to three different social events per week with each other. Consequently, this expanded group of social acquaintances supplies Greeks with a larger group of stimulant distributors.

With such a high demand for the product and its restrictive Schedule II classification, one may assume acquiring these stimulants would be expensive; it is not. Many interviewees claimed to get their stimulants for free from brothers or close friends with prescriptions. "Most of the time," explained Kenneth, "Brian [his fraternity brother] will just give me one of his extras when I need it. He doesn't care."

Most buyers and sellers generally agree stimulant sales are not about the money. "You can't really make money off your brothers," seller Mark explained. "I may ask for 5 bucks of something, sometimes. You know, just to cover costs, but you don't do it to make money or anything like that." Buyer, James, also sees these distributors as benevolent: "They do it to be cool. It is a nice thing to do for people if they don't abuse it. That's all. No big deal or anything. Just to be cool."

How and why do these diagnosed ADHD students have a surplus of these highly regulated pills? The answer, as Wayne told us, is "no one takes it every day. You really don't need it all the time. Like I only have classes on Tuesdays and Thursdays any more, so, unless I have a test, I just take it then." Robert, claims "it beats up your system." Common side effects mentioned, by both medical and nonmedical users, include the inability to sleep, sweating, increased heart rate, loss of appetite, and impotence. "To take it every day," explains David, "would really mess you up. You would like never eat or sleep well. You just can't do it, really."

Consequently, students with a prescription were strategic about stimulant use. Most never take their medication on weekends. Others take it only on the days they have classes, or when their work demands high levels of attention and concentration. Glen, a diagnosed ADHD student, explains: "I only take it when I need it. It has to be a busy day or something. Like a test. Or I forget to do something and I'm late." Thus, many prescribed students have a surplus of medication each month, and they see selling their leftovers as a morally inconsequential win/win decision. As James framed it, "What am I going to do with all those pills? So I figure, if I can help out some friends and make some beer money, life is good [laughs]."

Increased Use With Time

DeSantis et al. (2009) found an overall prevalence rate of nonmedical stimulant use among fraternity members of 55%. As alarming as this figure is, the

numbers of nonmedical users become even more shocking as these students progress toward graduation. Freshman fraternity members' use was at 35%, an already high number. This percentage increased, however, to 53% their Sophomore year, 62% their Junior year, and 86% their senior year (DeSantis et al., 2009).

Increased use over time may be explained by three factors. First, students perceived upper-division classes to be significantly more challenging and demanding. Consequently, the need to study longer, improve concentration, and increase cognition, three primary motivations for stimulant use (see section 2 for more detail), also increases. As John, a senior, explained, "Classes get really tough. It's not like I'm taking 100-level classes. I mean, I really, really need to work just to like make it in upper division."

Second, the more time students spend on campus, the larger their social network becomes; this is especially true in the Greek system. This larger social network exposes them to more students with prescriptions for ADHD medications, the primary distributors of stimulants on campus. Thus, it is easier to find and procure stimulants. Brain expands: "When I moved into the house [sophomore year], it was just always around. I think like five guys in the house had it. . . . Freshman year in the dorms, it was harder to find sometimes, you know, unless you knew a kid from high school."

Finally, the health risk (e.g., "drugs will kill you") and ineffectiveness (e.g., "Stimulants can't work on non-ADHD users") arguments used by students to justify abstention early in their college career may be rebutted over time by first-hand observations and second-hand testimony about the drug's safety and efficacy. "When I first got on campus," explained Kevin, "I was scared to try anything, you know, except beer. But you see it's no big deal. You know it's not like those movies where kids are freaking out or dying." For Mitch, testimonies from peers finally won him over. "After a while, after everyone is telling you how great it is, you think, why not. I mean, it really worked for that guy, so why not?"

DISCUSSION

The purpose of the current study was to gain a more in-depth understanding of nonmedical ADHD stimulant use by fraternity men on a college campus using qualitative methods. In order to augment the many quantitative studies on this topic (McCabe et al., 2005, 2006; Rabiner, Anastopoulos, Costello, Hoyle, & Swartzwelder, 2008; Shillington et al., 2006), the current study aimed to contextualize nonmedical stimulant use through qualitative methodologies. Results indicate that most users possessed limited knowledge of prescription stimulants, appropriate doses, physiological or psychological side effects, or legal consequences of nonmedical use. It has also been discovered that first-time use almost exclusively took place during periods of high academic stress. In particular, finals week was reported to produce a context of anxiety and

desperation that made students exceptionally vulnerable to the highly publicized promises of stimulants. Additionally, most students reported that procuring stimulants was an easy, unobtrusive act. In interview after interview, we were told not only how stimulants had become part of the campus culture, being openly and positively discussed in public settings, but how, as Robert, an undeclared freshman, confessed, “Anybody really can get it just about anytime they want it. It is easier than beer to get.”

The data also reveal a general perception among nonmedical users that ADHD medications enhance cognitive ability. In fact, statements from some interviewees almost advertise this drug as a miracle for any student striving for academic success. The irony, of course, is students with lower GPAs are the most likely candidates for use of ADHD stimulants (McCabe et al., 2005, 2006; Rabiner et al., 2008). This may suggest these students are seeking a “quick fix” to their academic troubles, something that may often *not* be realized.

In a related point, the current study did not attempt to measure if prescription stimulants actually improved non-ADHD students’ abilities in the classroom. The proliferation of research examining how ADHD medications affect cognitive functioning focuses on individuals diagnosed with ADHD. These studies show that ADHD prescription medications enhance cognitive attention (Hood, Braid, Rankin, & Isaacs, 2005), memory (Pietrzak, Mollica, Maruff, & Snyder, 2006; Wilson, Cox, Merkel, Moore, & Coghill, 2006), and other cognitive and behavioral functions (Mollica, Maruff, & Vance, 2004). Unfortunately, we know little about how these medications actually work on a physiological level among non-ADHD users. Future research should investigate claims of enhanced cognitive aptitude in order to better understand and assess such claims. Such research would be most useful if it took into account the context in which college students use such medications (e.g., during extended sleep-deprived periods). Indeed, health communication messages to reduce ADHD stimulant use would likely be improved by a clearer understanding of *actual* versus *perceived* effects of this prescription medication.

Perhaps the most revealing finding mined from our data was the general lack of guilt or dissonance expressed by many nonmedical users. Most viewed stimulant use as not only physically and psychologically harmless, but as morally acceptable since it was primarily used for academic purposes. Health behavior theories such as the Health Belief Model (Janz & Becker, 1984) and Extended Parallel Process Model (Witte, 1992) suggest that a pre-condition to taking protective action is perceiving that a clear threat exists. In absence of that, it is unlikely that students will, on their own, take protective action (i.e., cease their use of ADHD stimulants). In addition, the fact these stimulants are prescribed pharmaceuticals available at local pharmacies also appears to supply moral solace. In short, the nonmedical use of ADHD medication seems to be *stigma free* in the conceptions of the majority of our subjects. This finding does not bode well for traditional intervention efforts.

Implications for Interventions/Campaigns

What can be done to stem the tide of this growing problem of ADHD stimulant use in fraternities? There are a number of potential strategies that could be undertaken. These include:

1. targeting the student “suppliers” of the stimulants, since just a few suppliers make the drug accessible to so many students. It is clear that students with prescriptions for ADHD stimulants do not use their monthly allotment of pills, which creates a surplus of medications that are then given away or sold to students without prescriptions (DeSantis et al., 2008). Consequently, it is recommended that health care providers spend more time in consultation with patients regarding how many pills are actually needed each month. Pharmaceutical vendors should play a role in communicating with these health care providers about how their medications are being misused. If the surplus of pills can be reduced or eliminated, it would go a long way in reducing the potential illegal supply of ADHD stimulants;
2. educating nonmedical “consumers” of ADHD stimulants with regard to potential negative threat (i.e., negative consequences) of using the drug, perhaps through health communication campaigns (Derzon & Lipsey, 2002; Noar, 2006). Both our current and previous (DeSantis et al., 2008) research found that students engaged in nonmedical use of ADHD stimulants despite very little knowledge about the drug, its potential dangers and side effects. Educational campaigns could thus be beneficial in reducing demand for ADHD stimulants; and
3. increasing efforts by Universities and Greek organizations in providing accessible academic support for their members, particularly during periods of high academic stress. Multi-pronged strategies to reduce ADHD stimulant use may be warranted, particularly among the highest using populations (e.g., fraternity members, upperclassmen). Without such efforts, trends in ADHD stimulant use may continue to get worse over time.

Limitations and Future Directions for Research

The central limitation of the current research is the use of a convenience sample of fraternity members from a single university. Given this, there is no guarantee that these results generalize to all fraternity members or to other universities. Future research should include multiple universities in different regions of the United States, including public, private, large, and small campuses, as well as universities with varying admissions standards. A second limitation of the current research is that it only provides the perspective of the illegal user, not that of the supplier. Future studies would benefit from greater attention paid to the small number of individuals with prescriptions who appear to supply large proportions of college campuses with illegal ADHD stimulants. Of particular interest is how

these “distributors” end up with extra pills each month (i.e., their schedule of taking their medications), as well as how they morally justify the sharing of their medications with peers. A final limitation is that the current study treats all fraternities as essentially the same. Research in the date rape area, however, shows that there are particular factors (e.g., high school consumption, sexist behavior, dirty bathrooms) that differentiate “high risk” from “low risk” fraternities. Future research should attempt to connect features specific to particular types of fraternities to rates of use of ADHD stimulant medications.

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